

GROUP WORK PROJECT # _1__
GROUP NUMBER: ____5178____

MScFE 560: FINANCIAL MARKETS

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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Question1.:

Ans

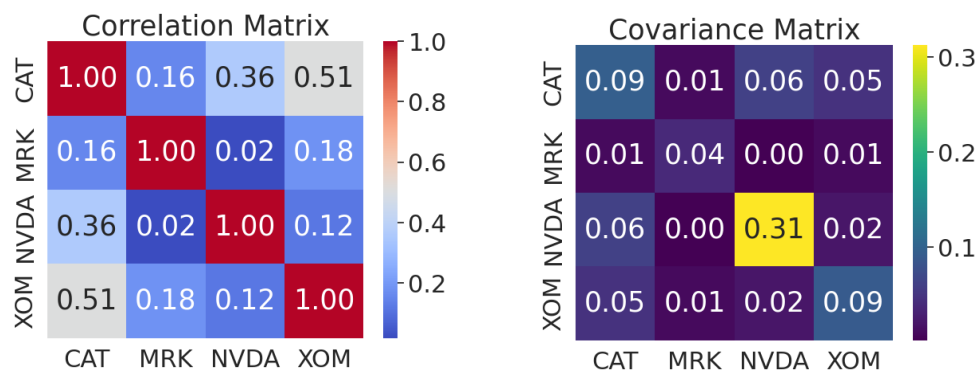
1.1 Portfolio with Income Stocks (Student: Salvatore Alessandro Chiruzzi)

1.1.1 Non-Optimized Portfolio

1. Caterpillar Inc. (CAT)
2. Merck & Co Inc (MRK)
3. Nvidia Corp (NVDA)
4. Exxon Mobil Corp (XOM)

We have selected four stocks from different sectors, which is important for minimizing risk and taking advantage of opportunities in different market conditions. This is well illustrated by the correlation and covariance matrix below.

	CAT	MRK	NVDA	XOM
Average Return	0.2603	0.2439	0.6147	0.2023
Volatility	0.3082	0.1939	0.5589	0.2994
Skewness	0.2991	0.1093	0.8073	-0.1974
Kurtosis	2.6998	1.3304	4.7660	0.9553



First of all we have downloaded 2 years of daily close prices using the Python tool “yfinance”, and calculated the daily percentage change. We have then annualized the daily statistics of return and volatility assuming 252 trading days in one year (i.e. multiplying the daily average and standard deviation by 252 and its square root respectively).

1.1.2 Optimized Portfolio (with weights)

We have used the Python package PyPortfolioOpt to find weights that maximize the Sharpe ratio. The tables with the statistics after optimization is reported below. For the optimization we used the methods

“EfficientFrontier” and “Discrete Allocation”. We have obtained a **Sarpe Ratio of 1.57**. Of course in this case we don’t report the correlation and covariance matrix because they would be the same as before.

The Efficient Frontier methods returned the following weights:
CAT **4.95%**, MRK **70.2%**, NVDA **19.9%**, XOM **4.95%**

	Optimized Portfolio
Average Return	0.3164
Volatility	0.1871
Skewness	0.1659
Kurtosis	1.2117

The Discrete Allocation method (based of a portfolio value of 10,000 USD) resulted in the following allocation: [CAT: 2, MRK: 58, NVDA: 3, XOM: 5], with 34.28 USD of remaining funds.

1.1.3 Shorting & Credit Risk for Income Stocks portfolio

- **Shorting**
 - This portfolio can be sold short. Anyway even if it is technically possible to short sell income stocks, doing so would be contradictory. Income stocks are usually chosen for their potential to provide a steady behavior over a long-term period offering returns with dividends. Conversely short-selling is a speculative strategy based on the assumption that a stock price will decrease significantly in the short term.
 - To short-sell this portfolio an individual should borrow shares of these companies from a broker and sell them on the open market at the current price. Later he/she should buy back the stocks (at a lower price if the shorting strategy proves to be successful) to return them to the broker
- **Credit Risk**
 - In a traditional sense of stocks (and so this portfolio) don’t have “credit risk” in the sense that bonds do, because buying a stock does not mean that an individual is lending money to a company expecting to be payed back after a certain period of time
 - Even if the portfolio itself is not exposed to credit risk, the companies (CAT, MRK, NVDA, XOM) could be affected by problems related to their ability to manage debt or paying dividends

1.2 Portfolio with Crypto-Currency (Student: Vikas Gautam)

1.2.1 Non-Optimized Portfolio

Asset class: Crypto-Currency

Assets chosen: Bitcoin (BTC), Ethereum (ETH), Tether (USDT), BNB

I have downloaded four years of daily Adj Close data (from 1 Jan 2019 to 7 Feb 2023) from yahoo finance and computed percent change as daily returns. Then computed annual return, variance and covariance by multiplying with 253 as there are 253 trading days in a year.

Summary Statistics and observation(s):

Asset	BTC-USD	ETH-USD	USDT-USD	BNB-USD
Average Annualized Return %	47.79%	70.96%	-0.16%	102.4%
Annualized Standard Deviation	0.59	0.762	0.0543	0.848
Sharpe Ratio	0.81	0.81	-0.03	1.21
Annual Covariance	0.253	0.506	0.00253	0.719
Variance	0.347	0.58	0.00295	0.72
Skewness	0.03	0.044	-0.67	2.98
Kurtosis	-0.38	-0.45	4.33	14.41

- high rise of Closing Prices for Bitcoin, BNB and Ethereum, particularly during 2021-2022, dropping further near 2023
- Fall for USDT Closing Prices after 2019-2020
- Average Sharpe Ratio for BTC, ETH and Optimal for BNB.
- high Returns for BNB
- BNB has high kurtosis (greater than 3)
- USDT is negatively skewed (long tail in the negative side) while BNB is positively skewed

Correlation Matrix for the Portfolio:

Corr Matrix	BTC-USD	ETH-USD	USDT-USD	BNB-USD
BTC-USD	1.00	0.82	-0.035	0.661
ETH-USD	0.82	1.00	-0.066	0.68
USDT-USD	-0.035	-0.066	1.00	-0.0626
BNB-USD	0.661	0.68	-0.0626	1.00

Variance and Covariance Matrix for the Portfolio:

- The diagonal values (e.g. [1,1]) of the table represent the variance, while the off-diagonal values represent the covariance between the two assets, e.g.: (row=1, column=2) value represents the covariance between BTC and ETH.
- These values of correlation matrix and covariance matrix is true for entire portfolio

Variance and Covariance Matrix	BTC-USD	ETH-USD	USDT-USD	BNB-USD
BTC-USD	0.348	0.37	-0.0011	0.33
ETH-USD	0.37	0.58	-0.003	0.44
USDT-USD	-0.011	-0.003	0.003	-0.003
BNB-USD	0.33	0.44	-0.003	0.72

- high positive correlation for ETH and BTC (0.82), slight negative or near zero correlation between USDT and (ETH, BNB) so movement of one will not affect other whereas correlation between ETH and BNB is moderate but positive (0.68).
- High variance noted for BNB-USD (0.72) and lower than moderate covariance between BNB and ETH (0.44) whereas covariance between USDT and (BTC, BNB) is slightly negative or near zero.

1.2.2 Optimized Portfolio (with weights)

Portfolio with different weight combinations as assets:

	Portfolio1 (Min. Volatility or Std. dev.)	Optimal Portfolio2:	Portfolio3: (medium return with medium volatility)	Portfolio4: (high Returns)	Portfolio5: (high returns)
Volatility or Annualized Std. dev. % :	19.27	73.3	48.25	72.46	78.55
Sharpe Ratio:	1.03	1.212	1.216	1.206	1.217
Annual Avg. Returns:	19.93%	88.83	58.7	87.41	95.52
Weight1 (BTC):	0.04	0.026	0.045	0.07	0.01
Weight2 (ETH):	0.18	0.25	0.094	0.25	0.14
Weight3 (USDT):	0.72	0.042	0.37	0.034	0.02
Weight4 (BNB):	0.05	0.69	0.49	0.65	0.83

- We have experimented on five hundred probable combinations of weights and picked the top ones for min volatility, max returns and max sharpe ratio.

- We can choose to have a more than moderate weightage to either one of BTC, USDT, or BNB, or ETH (i.e. near 50%) and can have good Sharpe ratio or good returns.
- More than 55% weight to USDT and close to 10% weight to ETH is likely to bring minimum volatility to portfolio
- Close to 50% of BNB combined with around 10% of BNB would likely to give max returns
- Combined aggregate weight of BNB and USDT with more than 75% would bring max Sharpe Ratio

1.2.3 Shorting and Credit Risk for Crypto-Currency Portfolio

Ques.: Can Portfolio having Bitcoin or Ethereum or any Cryptocurrency asset be Shorted?

For investors who believe that Bitcoin, Ethereum or Tether will crash at some point in the future, **Shorting the currency** could be a good option. For those who want to sell Crypto Asset short, i.e. make a profit by betting against the fall in Bitcoin or any crypto asset price. In short selling, we borrow the asset which is at higher price and sell it, later when the price falls, we buy the asset and return it to the lender and thus make profit on price difference. But it is risky too and the market is unregulated and may involve high fluctuations on both upside or downside. There are many investment options available:

1. Margin trading

By taking advantage of margin trading, we can borrow money from broker or lender and do trading by investing less cash from our side and reap benefits with higher percentage return on investment (ROI). It provides leverage and can result in high gains as well as high losses too.

2. Futures Market

There is a futures market for Bitcoin, as there is for any other crypto assets. If we sell a futures contract, would indicate a bearish prediction that the price of Bitcoin will fall. Thus, we can short Bitcoin etc.

3. Trading with binary options

We can also short Bitcoin with call and put options. If we want to short the crypto-currency, we can execute a put order, probably through an escrow service. This means that we would aim to sell the currency at today's price, even if the price falls later. One of the advantages is that we can limit our losses in any unfavorable condition by letting the option expire and we only lose the premium.

4. Prediction Markets:

Prediction markets, are another way to short Bitcoin. Prediction markets for cryptocurrencies are similar to mainstream markets. Investors can create an event and place a bet on the outcome of the event. So we could predict that Bitcoin will fall by a certain margin or percentage, and if someone takes our bet, we would profit when this happens. Popular crypto prediction markets include Augur, GnosisDAO and Polymarket.

5. As Inverse Exchange traded products

6. using Contract for Differences (CFDs) for any crypto asset

7. Short selling Crypto Assets or Tokens like Bitcoin Assets or tokens

Ques.: Does this portfolio have credit risk?

Credit risk is the risk of default on a debt that may arise if a borrower fails to meet its payment obligations. While many types of crypto-assets have no inherent credit risk, in practice they can **indirectly pose a credit risk**. For example, cryptos are used as a means of payment or as collateral in a significant

proportion of trading and transactions, and sudden price drops reduce their value. Accepting crypto assets as collateral with inadequate legal protection and haircuts can increase credit risk. The volatility of crypto assets could lead to negative wealth effects on the balance sheets of exposed companies or households, which in turn could reduce repayment capacity and deteriorate bank asset quality through an increase in NPLs.

The credit risk may not be significant in many cases, as some crypto assets have no claim on the issuer and are considered non-financial assets. For example, Bitcoin, which is classified as a non-financial asset, has no consideration. The risks associated with crypto issuers are particularly high if the crypto asset is linked to a claim. For e.g., the assets used by the issuers to hedge the cryptocurrencies issued are exposed to rapid changes in market prices, counterparty default or lack of access to liquidity.

Stablecoins such as Tether USD (USDT) are a kind of bridge between volatile crypto assets and very stable real assets. They offer users a higher degree of price stability than other cryptocurrencies. Stablecoins can have credit risk as they can have a claim on the issuer and there could be concerns about the quality of the assets.

Crypto companies may abandon traditional financial products after realizing large losses in their core crypto businesses. Also, there is credit risk in case of bankruptcy claim by a firm or fraud transactions or currency or token stolen from digital wallets or refusing to give the creditors their money back like happened with FTX.

Although some crypto-assets do not have a claim on an issuer, the **credit risk of the counterparty** may be relevant if the trade in cryptocurrency assets is significant. This particularly includes **cross-border repayment risk** on exports and imports. Risk triggers include sharp fluctuations in the prices of crypto assets or the underlying assets, corporate defaults.

Functions/Actors	Credit Risk	Market Risk	Liquidity Risk
Issuers	X	X	X
Miners/Validators		X	
Crypto exchanges/Platform	X (Risk of default of borrowers)	X	X
Wallet providers	X (Risk of default of asset holders)	X	X
Investors	X (Risk of default for claim based cryptos)	X	
Payment Providers	X	X	X

2. Question2: Interview transcript on FTX crisis

Ans.:

Student A: Since we are two persons in our group. Journalist(J1): shared by Vikas Gautam, and

Alessandro

Student B: Technical interviewee (TIE2): role play by Vikas Gautam and

Student C: Non-technical interviewee (NonTIE3): by Alessandro

Journalist1: First question, please tell me what happened to FTX?

NonTIE3: The rapid collapse of cryptocurrency exchange FTX, which stands for 'Futures Exchange', in 2022 has caused grave ripple effects for the industry, fueling public skepticism and upending cryptocurrency services that relied on it. FTX was the third-biggest cryptocurrency exchange at the time.

The collapse of FTX occurred within ten days in November 2022, triggered by a scoop from cryptocurrency news site CoinDesk which exposed that the majority of assets held by Alameda Research, a sister company, consisted of FTT and some other tokens created and controlled by FTX and its insiders, rather than to be a cryptocurrency with a market-driven and proven value. During FTX's rapid growth phase, the company skipped the important step of establishing a corporate governance structure and controls were never put in place.

Several investors and users withdrew their money from FTX, causing the exchange to go bankrupt and become insolvent. The revelations caused a stir, raising concerns about FTX's disproportionate indebtedness to Alameda Research, and its business practices on shaky financial accounting metrics, and the associated risks to financial management. FTX's balance sheet was published, revealing a lack of diversification and an over-reliance on the two companies. FTX crashed due to mismanagement of funds, lack of liquidity and the large volume of withdrawals.

Findings included security flaws such as private keys for moving *crypto* assets often being left in unencrypted files and crypto assets being left unsecured in Internet-connected hot wallets. Also cited were failures to use multifactor authentication in critical areas and multiple individuals with access to the billions of dollars in crypto wallets.

Shortly after declaring bankruptcy, FTX asserted that about \$500 million worth of assets might have been pilfered via "unauthorized transactions." Elliptic, a cryptocurrency compliance provider that detected the withdrawals, reported that the hacker used a technique known as "on-chain spoofing" to empty the wallets over a period of several days. According to reports, the hacker used the money to purchase ether (ETH), a cryptocurrency.

As Founder and CEO, Sam Bankman-Fried have found out, "you can fool some of the people some of the time, but you can't fool all of the people all of the time". Even with a subject as seemingly confusing as

crypto. SBF, as he is commonly known, could theoretically face more than 50 years in prison if he is convicted in 2024. This case illustrates ethical issues, his deceitful practices, his lack of respect for the truth and his disregard for interpersonal trust, that are applicable even beyond the world of cryptocurrencies and blockchain.

Journalist1: What is the future of FTX and Consequences of Collapse?

TechIE2: Years will pass while investors, regulators, and courts attempt to recover money that was lost to these firms and stop a disaster from occurring again. FTX, which was the biggest exchange collapse in the brief history of cryptocurrencies, cautious investors have now refrained from investing in the market

After FTX collapsed, a wave of customer withdrawals caused cryptocurrency exchanges such as crypto.com to shrink. Cryptocurrency lenders and banks are going through bankruptcy liquidations and facing legal action for approving loans to FTX and Alameda Research that were grossly undercollateralized and for ignoring the risk profile of the repayments.

Customers and investors could not get their money back in full from FTX. Depending on how much money the bankruptcy trustee may recover, creditor claims may be valued at a substantial discount to their principal and are subject to a drawn-out bankruptcy estate procedure. A few of these actions like tracking down the whereabouts of cryptocurrency, requesting politicians to filing a lawsuit to recover assets from corporate partners.

J: PLs tell on Regulatory Crackdown?

NonTE3: Authorities have demanded more stringent government regulation of cryptocurrencies. Law enforcement has attempted to restrict exposure to traditional markets and has increased oversight of cryptocurrencies on a national and worldwide level. Congressmen have stated that they are more likely to pass new laws establishing regulations for digital currencies and exchanges. In the crypto realm, there is a need to improve both the sophistication and clarity of rules. The US government has just fined Binance – the world’s largest crypto exchange – 4.3 billion US dollars (3.4 billion pounds) for being involved in money laundering.

Cryptocurrency exchanges such as BTC Markets, Kraken, Swyftx and Coinbase are keen for Australia to introduce regulation for the controversial crypto space to protect investors and restore confidence in an industry that was badly damaged by the implosion of FTX. The European Commission Regulation on markets in crypto-assets ("MiCA") is part of the broader "Digital Finance Package", which aims to develop a digital finance strategy that will enable the EU to manage the digital revolution in the financial sector

J: PLs tell something about how this works?

TechIE2: Traders and investors can buy and sell using cryptocurrencies without going via banks or brokerages. Peer-to-peer cryptocurrency transactions on platforms like FTX and its competitor Binance are made possible by blockchain technology, which eliminates the need for the middlemen. A centralized system exists to match buyers and sellers, with market makers providing liquidity and charging dealers for each deal.

Thanks to its centralized mechanism, FTX was able to lend money to struggling cryptocurrency companies. In order to complete the books of its sister firm, it also used exchange-issued tokens, or FTT. This raised the possibility of being exposed to a market meltdown.

However, a newer approach called decentralized exchanges functions differently in terms of governance and cryptocurrency pricing, which may lower these concerns. They make it possible for investors to purchase and sell tokens at a price set by an algorithm. Instead of using paid market makers.

Because decentralized exchanges are non-custodial, individual investors have complete access to their cryptocurrency wallet balances and are free to quit trading or withdraw funds at any moment without worrying about the exchange freezing their assets.

However, one drawback of decentralized exchanges over centralized models like FTX and Binance is that traders may only trade distinct cryptocurrencies on the exchange; they are unable to exchange fiat, or traditional currencies issued by governments or nations. A trader may encounter difficulties executing their intended transaction if the liquidity pool is too tiny, as the amount of each deal is contingent upon it. Given the failure of centralized financial institutions (CeFi) such as FTX, proponents argue that decentralized financial institutions (DeFi) are the future: only “true” DeFi can be resilient.

J: Please tell something more. What else they can do?

TechIE2: The FTX scandal shows that several elements are necessary to prevent fraud and promote ethics:

- Closer monitoring of crypto assets: Rather than depending on annual reports that sometimes contain glaring mistakes, one approach should be to avoid a repeat of the FTX debacle and to monitor crypto exchange assets in real time. 'Proof of reserves' can be supplied by an impartial third party. This indicates that the company releases audit reports to offer a third-party examination of an exchange's balance sheet, monitoring the movement of funds into and out of investors' exchange wallets. This would reveal any unexpected activity that could lead to systemic breakdowns, like the use of exchange reserves as collateral for loans to cryptocurrency companies.
- Improved crypto risk assessments: Financial authorities must also implement a suitable methodology for assessing cryptocurrency risk. Independent audits and stress testing of on-chain data pertaining to transactions on a blockchain network. It may be possible to implement regulations that limit the use of an exchange's tokens for lending to cryptocurrency companies. Increased consumer protection may also stop exchanges from stopping withdrawals, which would deny traders access to funds held by a faltering exchange. Cryptocurrency is still relevant even in the midst of the "crypto winter." New models and appropriate rules could support the industry's growth and recovery, and they might even encourage more mainstream financial markets to adopt decentralized finance.
- Independent auditors,
- A strong general counsel with experience in a regulated financial institution,

- An internal compliance department that reviews the activities of all employees with trading responsibilities, and
- A strong financial crime or compliance controls group that looks for conflicts of interest, sanctions, political involvement or influence, anti-money laundering, know-your-customers, risk management, etc.

J: Could FTX Make a Comeback?

NonTIE3: Sometimes companies that file for bankruptcy are bought up and reopened. The trade secrets, intellectual property and brand recognition of the target can be valuable to the buyers. But commercial arrangements can change, and the success of the relaunched business is not assured. It is difficult to predict the factors affecting these outcomes, which range from macroeconomic circumstances to management skills and customer loyalty. The collapse of FTX and the resulting fallout has made the cryptocurrency sector a focal point for research into current financial crime and corporate compliance.

Thanks.

Reference:

Bitcoin USD 3-year historic price link:

<https://finance.yahoo.com/quote/BTC-USD/history?period1=1675209600&period2=1675987200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>

Ethereum USD 3-year historic price link:

<https://finance.yahoo.com/quote/ETH-USD/history?period1=1675209600&period2=1675987200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true> <https://finance.yahoo.com/quote/ETH-USD/history?period1=1675209600&period2=1675987200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>

Tether USD (USDT) 3-year historic price link:

<https://finance.yahoo.com/quote/USDT-USD/history?period1=1675209600&period2=1675987200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>

BNB USD 3-year historic price link:

<https://finance.yahoo.com/quote/BNB-USD/history?period1=1675209600&period2=1675987200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>

Online links on FTX crisis:

[FTX scam explained: Everything you need to know \(techtargget.com\)](#)

[The crypto industry is still reeling from the effects of FTX \(consultancy.uk\)](#)

[FTX and Binance: how latest crypto scandals could influence public opinion on digital currency regulation \(yahoo.com\)](#)

[BTC Markets, Kraken, Swyftx, Coinbase among cryptocurrency exchanging hoping for new Australian regulations to prevent another FTX \(afr.com\)](#)

[Ethical Failure Led to the FTX Scandal | IMA \(sfmagazine.com\)](#)

[Case Study: FTX and Sam Bankman-Fried - Seven Pillars Institute](#)

[Crypto assets: a lesson from history – Deutsche Bank \(db.com\)](#)